

ZELIN (RICHARD) ZHU

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PROFILE

Self-directed AI builder and researcher: 48 public repositories spanning Claude-native community platforms, agent-safety tooling, and autonomous quantitative research. Lead a 3-person university research team (manuscript in preparation for ICLR 2027); organized AI-literacy programming for 60+ community members. Bilingual (English & Mandarin).

EDUCATION

Chapel Hill High School — Class of 2027, GPA: Summa Cum Laude *Chapel Hill, NC | Aug 2023 – May 2027*

Coursework: AP Calculus BC, AP Statistics with Calculus, AP Physics C: Mechanics, AP Physics C: E&M; Multivariable Calculus & Linear Algebra (self-directed, MIT OCW)

Durham Technical Community College — Dual enrollment, GPA: 4.00 *Durham, NC | Aug 2025 – May 2027*

University of North Carolina at Chapel Hill — Concurrent coursework while in high school *Chapel Hill, NC | Jul 2026 – May 2027*

Y Combinator Startup School 2026 — Selected attendee, hand-picked builder cohort *San Francisco, CA | Jul 2026*

Selected from ~50,000 applicants (top ~5%); among the youngest admitted, at 17.

Certifications — **Anthropic Academy:** Claude 101 · Claude Code 101 · AI Fluency: Framework & Foundations · AI Fluency for Nonprofits · Introduction to Claude Cowork · Teaching AI Fluency · Introduction to Model Context Protocol

TECHNICAL SKILLS

Languages: Python, C++, C, Java, TypeScript/JavaScript, SQL, HTML/CSS

Claude & agentic engineering: Claude API & Claude Code — multi-agent orchestration, skills authoring, loop engineering, MCP; Google ADK; retrieval-grounded generation; agent auditing & guardrail design

AI/ML: PyTorch, scikit-learn, TensorFlow, Hugging Face Transformers, pandas, NumPy; time-series forecasting (TFT, N-BEATS, MAML), Mixture-of-Experts with regime-aware gating, SVM, PCA; ASR benchmarking (Whisper — WER/CER, disparity analysis)

Quant research: Backtesting & signal generation (ccxt), multiple-testing discipline (pre-registered criteria, deflated Sharpe), Monte Carlo simulation, ensemble modeling (Bradley-Terry, Pythagorean expectation)

Web & tools: React, Next.js, Node.js, React Flow, Prisma/Drizzle, Tailwind, Docker, Git/GitHub, Vercel, Jupyter

EXPERIENCE

UNC ACM Lab | *Lead Researcher* *Chapel Hill, NC | Jun 2025 – Present*

- Lead a 3-person team researching multi-agent AI systems through a cognitive-science lens; manuscript in preparation for submission to ICLR 2027.

UNC PRIMES | *Scholar* *Chapel Hill, NC | May 2026 – Present*

- Selected for UNC's Program for Research in Image Processing and Machine Learning for Emerging Scholars; conducted machine-learning and computer-vision research and was invited to continue with the lab beyond the program.

MeridianAlgo | *Quantitative Developer* *Remote | Sep 2025 – Present*

- Build and maintain Apex Analysis, an open-source ML stock-analysis and research platform, on a 5-person engineering team (meridianalgo.org).
- Develop model-training pipelines for equity-market prediction and signal generation (Python, scikit-learn).

AI9ha (Nexus Tensor Alpha) | *Founder* *Chapel Hill, NC | Nov 2025 – Present*

- Building an AI-native quantitative research system: Mixture-of-Experts forecasting (TFT, N-BEATS, MAML) with regime-aware gating and a multi-LLM agent layer for unstructured signal processing.
- Designed for evidence-backed equity research with review-bound proposals and approval-gated paper trading — a human signs off before anything executes.

SELECTED PROJECTS

RTP Pathway — *Claude-grounded opportunity platform for Triangle-area students* *TypeScript, Next.js, Python*

- Built a community platform where local high-schoolers discover verified research programs, internships, and competitions; Python/pandas pipeline deduplicates, validates, and routes listings through a human review queue.

- Natural-language search answered by Claude and grounded exclusively in the verified database, with per-answer source attribution and “last verified” dates; anonymous analytics surface zero-result queries to guide curation.

Scutum — *audit layer for coding agents* (live: scutum-orpin.vercel.app) *TypeScript, Next.js, Prisma*

- MVP of a black-box audit layer: records agent executions, risk-scores every tool call, blocks dangerous actions (secret exfiltration, unauthorized network access), and queues borderline actions for human approval.
- Replayable incident timelines and a policy editor with attack tests; demo replays a malicious GitHub-issue prompt injection being caught and blocked.

Quantlab — *autonomous research lab for falsifying trading hypotheses* *Python, Docker*

- Five-stage agent loop (hypothesize → implement → backtest → referee → memo) that generates and stress-tests crypto trading hypotheses; built with Claude Code as a self-pacing autonomous loop (loop engineering).
- Engineered against false discoveries: kill criteria pre-registered in git before results exist, frozen read-only backtest engine, an independent referee model applying zero-discretion rules, deflated-Sharpe multiple-testing correction, and a physically absent holdout period.

EqualVoice — *voice-AI equity benchmark* *Duke SCAI Hackathon — Finalist*

- Captained an all-high-school team to finalist against undergraduate, graduate, and PhD teams; benchmarked Whisper ASR across accents using Mozilla Common Voice.
- Measured 2–6× higher transcription-error rates and 15–25-point intent-classification gaps for non-native English speakers; text normalization recovered 10–15 points of intent accuracy.

Zhu Academy — *AI-powered learning platform* *React 18, Vite, React Flow, Claude API*

- Knowledge-graph tool that maps concepts into adaptive learning pathways.

Research Email Agent — *Kaggle capstone, open-sourced* *Python, Google ADK*

- Multi-agent outreach pipeline with Research, Profile, EmailWriter, and Validator agents for automated academic outreach.

Wharton HSDSC — *Team Captain / Modeling Lead* *Python*

- Multi-model ensemble (Pythagorean Expectation + Bradley-Terry) for ice-hockey simulation analysis.

LEADERSHIP & COMMUNITY

- **Founder**, UNA-USA Chapter at CHHS (SDGs 8, 9, 13) — organized the Durham AI Literacy Workshop: 60+ attendees, sponsored by Meta and Perplexity.
- **President**, AI Club, CS Club, and Research & Innovation Club, Chapel Hill High School.
- **Advisory Board Member**, Morehead Planetarium Teen Science Café and Museum of Life and Science Teen Advisory Board.

HONORS & AWARDS

- **USACO Gold Division**
- **Duke SCAI Hackathon** — Finalist, 2026
- **Georgia Tech Statistics and Probability Competition** — advanced to the top-40 second round; captain of the first-place winning team
- **Wharton High School Data Science & Investment Competitions** — Team Captain